Vaidehi Gohil

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EDUCATION

M.S. in Electrical and Computer Engineering - Northeastern University, Boston, MA | GPA: 3.5 May 2027
Relevant Courses: Embedded Systems in Deep Learning, Wireless Sensors & Internet of Things

B.E. in Electronics & Telecommunication – Vidyavardhini's College, Mumbai, India June 2024
Relevant Courses: Microcontrollers, Wireless Communication System, Computer Networks
Indian Patent Application Published, No. 202521062488 for Telehealth: Connectivity to Wellness

TECHNICAL SKILLS

Communication Protocols: I^2C, SPI, UART, RS-485, Modbus (RTU/TCP), BLE, OPC UA
Embedded: ESP-WROOM32, Raspberry Pi (Pico, W, 4, 5), Arduino, Nordic nRF7002, MSP430
Programming: C, C++, Python, Linux (shell, systemd), Git, OpenWRT, MQTT, HTTP services
Circuits & Components: MOSFET Power-Path Switching, Optocoupler Isolation Circuits, Sensor Interfacing

WORK EXPERIENCE

Graduate Research Assistant – Shepherd's Lab, Northeastern University | Github June 2026 – Present

- Added a new IMU driver to the open-source moteus servo firmware so the hip exoskeleton could read the MPU5060 IMU sensors, which the existing firmware did not support
- Building custom battery packs for a back exoskeleton and integrating the same IMU firmware across the back and hip exos

IoT Engineer – Faclon Labs, Mumbai, India November 2024 – August 2025

- Built fault-tolerant Python automation and telemetry services on OpenWrt gateways (timeout validation, retry/backoff, watchdog auto-recovery); deployed across 5+ remote industrial sites and authored setup and troubleshooting documentation used by field engineers
- Integrated heterogeneous field devices over I^2C, SPI, UART, RS-485, Modbus (RTU/TCP), and BLE; defined register maps and polling schedules; standardized outputs as structured JSON via MQTT/HTTP for downstream dashboards and faster field diagnostics
- Validated an in-house Battery Management System (BMS) PCB power-path stage (three MOSFETs); recreated the switching logic on a test rig with LED indicators to verify battery-to-external-supply state transitions during system bring-up

PROJECTS

Herald: The Assistant | EECE 7398 | Github April 2026 – May 2026

- Co-built a fully local, no-cloud voice control system; owned the server-side AI pipeline on a Raspberry Pi 5 with speech recognition (Whisper), on-device LLM inference (Llama 3.2 with quantized GGUF models via llama-cpp-python), text-to-speech (Kokoro), and an MCP-inspired tool dispatcher routing structured commands to ESP32 actuator nodes over MQTT
- Set up the headless deployment environment (systemd auto-boot, Wi-Fi hotspot, Mosquitto broker) and benchmarked end-to-end pipeline latency across the ASR to LLM dispatch to TTS path

Telehealth: Connectivity to Wellness | Patent No. 202521062488 | Mumbai, India August 2023 – January 2024

- Designed and prototyped a telehealth kiosk for rural deployment, integrating ECG, SpO_2, and temperature sensors for primary checkups and remote consultation
- Implemented a two-tier hardware architecture with ESP-WROOM32 reading sensor data and managing calibration, and a Raspberry Pi running a web dashboard for real-time visualization & video consultation
- Developed sensor drivers and data aggregation code for stable, reliable readings; documented system design, calibration, and usage procedures for future field deployment